

## Output

```
--- compiler:3.13.0:compile (default-compile) @ Polimorph ---
Recompiling the module because of changed source code.
Compiling 6 source files with javac [debug release 24] to target\classes

--- exec:3.1.0:exec (default-cli) @ Polimorph ---
Warna semut : merah
Posisi semut : X = 70
               Y = 90
Posisi semut : X = 80
               Y = 100

Warna lebah : Oranye
Posisi lebah : X = 30
               Y = 40
               Z = 50
Posisi lebah : X = 45
               Y = 55
               Z = 65
-----
```

```
11 public class Polimorph {
12
13     public static void main(String[] args) {
14         Koordinat ks1 = new Koordinat(70,90);
15         Koordinat ks2 = new Koordinat(80,100);
16         Koordinat3D kl1 = new Koordinat3D(30,40,50);
17         Koordinat3D kl2 = new Koordinat3D(45,55,65);
18         Semut s = new Semut("merah", ks1);
19         Lebah l = new Lebah("Oranye",kl1);
20         System.out.println("Warna semut : " + s.getWarna());
21         s.info();
22         s.gerak(ks2);
23         s.info();
24         System.out.println("");
25         System.out.println("Warna lebah : " + l.getWarna());
26         l.info();
27         l.gerak(kl2);
28         l.info();
29     }
30 }
31
```

```
12 public class Koordinat {
13     private int x, y;
14     public Koordinat(int x, int y) {
15         this.setY(y);
16         this.setX(x);
17     }
18     /**
19      * @return the x
20      */
21     public int getX() {
22         return x;
23     }
24     /**
25      * @param x the x to set
26      */
27     public void setX(int x) {
28         this.x = x;
29     }
30     /**
31      * @return the y
32      */
33     public int getY() {
34         return y;
35     }
36     /**
37      * @param y the y to set
38      */
39     public void setY(int y) {
40         this.y = y;
41     }
42 }
43
44
45
46
47
```

```
1 public abstract class Serangga {
12     private String warna;
13     private Koordinat posisi;
14     public Serangga(String warna, Koordinat k){
15         this.posisi = k;
16         this.warna = warna;
17     }
18
19     /**
20      * @return the warna
21      */
22     public String getWarna() {
23         return warna;
24     }
25
26     /**
27      * @param warna the warna to set
28      */
29     public void setWarna(String warna) {
30         this.warna = warna;
31     }
32
33     /**
34      * @return the posisi
35      */
36     public Koordinat getPosisi() {
37         return posisi;
38     }
39     public void gerak(Koordinat k){
40         this.posisi = k;
41     }
42     public abstract void info();
43 }
44
```

```
11 public class Koordinat3D extends Koordinat {
12     private int z;
13
14     public Koordinat3D(int x, int y, int z) {
15         super(x, y);
16         this.setZ(z);
17     }
18
19     /**
20      * @return the z
21      */
22     public int getZ() {
23         return z;
24     }
25
26     /**
27      * @param z the z to set
28      */
29     public void setZ(int z) {
30         this.z = z;
31     }
32
33 }
34
35
36 public class Semut extends Serangga{
37     public Semut(String warna, Koordinat k) {
38         super(warna, k);
39     }
40
41     @Override
42     public void info(){
43         System.out.println("Posisi semut : X = " + this.getPosisi().getX());
44         System.out.println("                Y = " + this.getPosisi().getY());
45     }
46 }
```

```
11 public class Lebah extends Serangga{
12
13     public Lebah(String warna, Koordinat3D k) {
14         super(warna, k);
15     }
16     public void gerak(Koordinat3D k){
17         // Memanggil gerak() dari Serangga (parent class)
18         super.gerak(k);
19     }
20
21     @Override
22     public void info(){
23         Koordinat3D pos3D = (Koordinat3D) this.getPosisi();
24         System.out.println("Posisi lebah : X = " + pos3D.getX());
25         System.out.println("                Y = " + pos3D.getY());
26         System.out.println("                Z = " + pos3D.getZ());
27     }
28
29 }
30
```